CS-GY 6033
Design and Analysis of Algorithms I

Lecture 5 | Part 1

Warmup

Exercise

- How fast can we query/insert with these data structures?

	Query	Insert
Unsorted linked list	$\Theta(n)$	$\Theta(1)$
Unsorted array	$\Theta(n)$	$\Theta(n)$
Sorted array	$\Theta(\log n)$	$\Theta(\log n + n) = \Theta(n)$
BST	$\Theta(h)$	$\Theta(h)$
?	$\Theta(1)$	$\Theta(1)$

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Lecture 5 | Part 2

Direct Address Tables

Counting Frequencies

- How many times does each age appear?

PID	Name	Age
A1843	Wan	24
A8293	Deveron	22
A9821	Vinod	41
A8172	Aleix	17
A2882	Kayden	4
A1829	Raghu	51
A9772	Cui	48
⋮	⋮	⋮

Exercise

What data structure would you use to store the age counts?

Direct Address Tables

- Idea: keep an **array** `arr` of length, say, 125.
- Initialize to zero.
- If we see age x, increment `arr[x]` by one.

Building the Table

```
# loading the table
table = np.zeros(125)

for age in ages:
    table[age] += 1
```

- Time complexity if there are *n* people? $\Theta(n)$

Query

```
# query: how many people are 55?
print(table[55])
```

$\Theta(1)$

- Time complexity if there are n people?

Counting Names

- How many times does each name appear?

PID	Name	Age
A1843	Wan	24
A8293	Deveron	22
A9821	Vinod	41
A8172	Aleix	17
A2882	Kayden	4
A1829	Raghu	51
A9772	Cui	48
⋮	⋮	⋮

Downsides

- DATs are **fast**.
- What are the downsides of DATs?
- Could we use a DAT to store:
  - zip codes?
  - phone numbers?
  - credit card numbers?
  - names?

Downsides

- Things being stored must be integers, or convertible to integers
  - why? valid array indices
- Must come from a small range of possibilities
  - why? memory usage. example: phone numbers

Hash Tables

- Insight: anything can be “converted” to an integer through **hashing**.
- But not uniquely!
- Hash tables have many of the same advantages as DATs, but work more generally.

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Lecture 5 | Part 3

Hashing

Hashing

- One of the most important ideas in CS.

- Tons of uses:
  - Verifying message integrity.
  - Fast queries on a large data set.
  - Identify if file has changed in version control.

Hash Function

- A **hash function** takes a (large) object and returns a (smaller) “fingerprint” of that object.
- Usually the fingerprint is a number, guaranteed to be in some range.

How?

- Looking at certain bits, combining them in ways that *look* random (but aren't!)

Hash Function Properties

- Hashing same thing twice returns the same hash.
- Unlikely that different things have same fingerprint.
  - But not impossible!

Collisions

- Hash functions map objects to numbers in a defined range.
  - Example: given image, return number in $[0, 1, 2, \ldots, 1024]$
- There will be two images with the same hash.
  - **Pigeonhole principle**: if there are n pigeons, $< n$ holes, there will a hole with ≥ 2 pigeons.
- **Collision**: two objects have the same hash

“Good” Hash Functions

- A good hash function tries to minimize collisions.

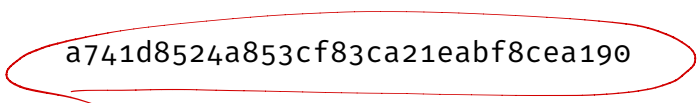
Hashing in Python

- The `hash` function computes a hash.

```
>>> hash("This is a test")
-6704585799574772O3
>>> hash("This is a test")
-6704585799574772O3
>>> hash("This is a test!")
1860306055874153109
```

MD5

- MD5 is a **cryptographic** hash function.
  - Hard to “reverse engineer” input from hash.
- Returns a *really large* number in hex.



- Used to “fingerprint” whole files.

Example

```
> echo "My name is Akbar" | md5
a741d8524a853cf83ca21eabf8cea190
> echo "My name is Akbar" | md5
a741d8524a853cf83ca21eabf8cea190
> echo "My name is Akbar!" | md5
f11eed2391bbd0a5a2355397c089fafd
```

Example

```
> md5 slides.pdf
e3fd4370fda30ceb978390004e07b9df
```

Why?

- I release a piece of software.
- I host it on Google Drive.
- Someone (Google, US Gov., etc.) decides to insert extra code into software to spy on users.
- You have no way of knowing.

Why?

- I release a piece of software & **publish the hash**.
- I host it on Google Drive.
- Someone inserts extra code.
- You download the software and hash it. If hash is different, you know the file has been changed!

Hashing for us

- Don’t need to know much about *how* the hash function works.
- But should know how they are used.

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Lecture 5 | Part 4

Hash Tables

Membership Queries

- **Given**: a collection of n numbers and a target t.
- **Find**: determine if t is in the collection.

Goal

- DATs are fast, but won't work for things that aren't numbers in a small range.
- Idea: hash objects to numbers in a small range, use a DAT.
- But must deal with collisions.

Hash Tables

- Pick a table size m.
  - Usually $m \approx$ number of things you'll be storing.
- Create hash function to turn input into a number in $\{0, 1, \ldots, m - 1\}$.
- Create DAT with m bins.

Example

```
hash('hello') == 3
hash('data') == 0
hash('science') == 4
```

Collisions

- The **universe** is the set of all possible inputs.
- This is usually much larger than m (even infinite).
- Not possible to assign each input to a unique bin.
- If `hash(a) == hash(b)`, there is a **collision**.

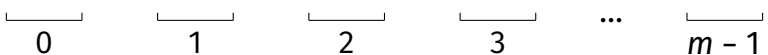
a and b are in the universe.

Example

```
hash('hello') == 3
hash('data') == 0
hash('san diego') == 3
```

Chaining

- Collisions stored in same bin, in linked list.
- **Query**: Hash to find bin, then linear search.



The Idea

- A good hash function will utilize all bins evenly.
  - Looks like uniform random distribution.
- If $m \approx n$, then only a few elements in each bin.
- As we add more elements, we need to add bins.

Average Case

- n elements in table.
- m bins.
- Assume elements placed randomly in bins[1].
- Expected bin size: n/m

[1] Of course, they are placed deterministically.

Analysis

- Query:
  - Time to find correct bin: $\Theta(1)$
  - Expected number of elements in the bin: n/m
  - Time to perform linear search: $\Theta(n/m)$
  - Total: $\Theta(1 + n/m)$
  - We usually guarantee $m = \Theta(n)$
  - Expected time: $\Theta(1)$.

Worst Case

- Everything hashed to same bin.
  - Really unlikely!

- Query:
  - $\Theta(1)$ to find bin
  - $\Theta(n)$ for linear search.
  - Total: $\Theta(n)$.

Exercise

What is the worst case time complexity of inserting an element into a hash table that uses chaining with linked lists?

$\Theta(1)$

Growing the Hash Table

- Insertions take $\Theta(1)$ **unless** the hash table needs to grow.
- We need to ensure that $m \leq c \cdot n$.
  - Otherwise, too many collisions.
- If we add a bunch of elements, we'll need to increase m.
- Increasing m means allocating a new array, $\Theta(m) = \Theta(n)$ time.

Main Idea

Hash tables support constant (expected) time insertion and membership queries.

Dictionaries

- Hash tables can also be used to store (key, value) pairs.
- Often called **dictionaries** or **associative arrays**.

Hashing in Python

- `dict` and `set` implement hash tables.
- Querying is done using `in`:

```
>>> # make a set
>>> L = {3, 6, -2, 1, 7, 12}
>>> 4 in L # Theta(1)
False
>>> 7 in L # Theta(1)
True
```

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Lecture 5 | Part 5

Fast Algorithms with Hash Tables

Faster Algorithms

- Hashing is a super common trick.
- The “best” solution to interview problems often involves hashing.

Example 1: The Movie Problem

- You're on a flight that will last *D* minutes.
- You want to pick two movies to watch.
- Find two whose durations sum to **exactly** *D*.

Recall: Previous Solutions

- Brute force: $\Theta(n^2)$.
- Sort, use sorted structure: $\Theta(n \log n) + \Theta(n)$.
- Theoretical lower bound: $\Omega(n)$?
- Can we speed this up with hash tables?

Idea

- To use hash tables, we want to frame problem as a **membership query**.

Example

- Suppose flight is 360 minutes long.
- Suppose first movie is fixed: 120 minutes.
- Is there a movie lasting (360 - 120) = 140 minutes?

```python
def optimize_entertainment_hash(times, D):
    hash_table = dict()
    for i, time in enumerate(times):
        hash_table[time] = i

    for i, time in enumerate(times):
        target = D - time
        if target in hash_table:
            return i, hash_table[target]
```

$\Theta(n)$

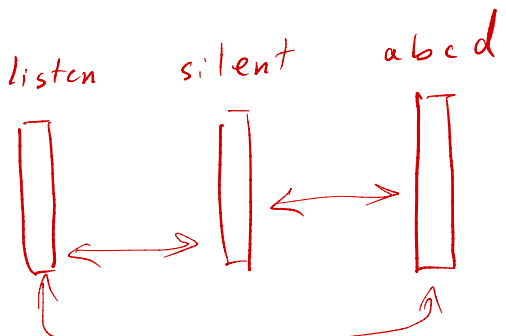
Example 2: Anagrams

Definition

Two strings `w_1` and `w_2` are **anagrams** if the letters of `w_1` one can be permuted to make `w_2`.

Examples

- abcd / dbca
- listen / silent
- sandiego / doginsea



Problem

- Given a collection of n strings, determine if any two of them are anagrams.

Exercise

Design an efficient algorithm for solving this problem. What is its time complexity?

Solution

- We need to turn this into a **membership query**.
- **Trick**: two strings are anagrams iff

```
sorted(w_1) == sorted(w_2)
```

```python
def any_anagrams(words):
    seen = set()
    for word in words:
        w = sorted(word)
        if w in seen
            return True
        else:
            seen.add(w)
```

words [abcd | dcab | ...]

seen = { }

w = "abcd"

seen = { "abcd" }

w = sorted("dcab")

w = "abcd"

$\Theta(n \cdot K \log K)$

my : $\Theta(n \cdot K)$

Hashing Downsides

- Problem must involve **membership query**.

Example: The Movie Problem

- You're on a flight that will last D minutes.
- You want to pick two movies to watch.
- Find two whose added durations is **closest** to D.

Hashing Downsides

- No locality: similar items map to different bins.
- There is no way to quickly query entry closest to given input.

Example: Number of Elements

- Given a collection of n numbers and two endpoints, a and b, determine how many of the numbers are contained in $[a, b]$.
- Not a membership query.
- Idea: **sort** and use modified `binary search`.

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Hash Table Drawbacks

Hashing Downsides

- No locality: similar items map to different bins.
- But we often want similar items at the same time.
- Results in many **cache misses**, **slow**.

Hashing Downsides

- Memory overhead.

Hash Tables vs. BSTs

- Hash Table: $\Theta(1)$ insertion, query (expected time).
- BST: $\Theta(\log n)$ insertion, query (if balanced).
- Why ever use a BST?

Hash Tables vs. BSTs

- Hash tables keep items in arbitrary order.
- Example: how many elements are in the interval $[3, 23]$?
- Example: what is the min/max/median?
- BSTs win when order is important.

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Lecture 5 | Part 7

Approximate Nearest Neighbors (ANN)

ANN

- **Given**: A set of points and a query point, *p*.
- **Return**: An **approximate nearest neighbor**.

Today

- **Approximate nearest neighbors** via **Locality Sensitive Hashing (LSH)**.

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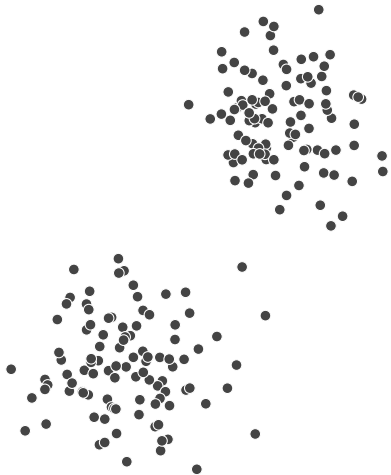
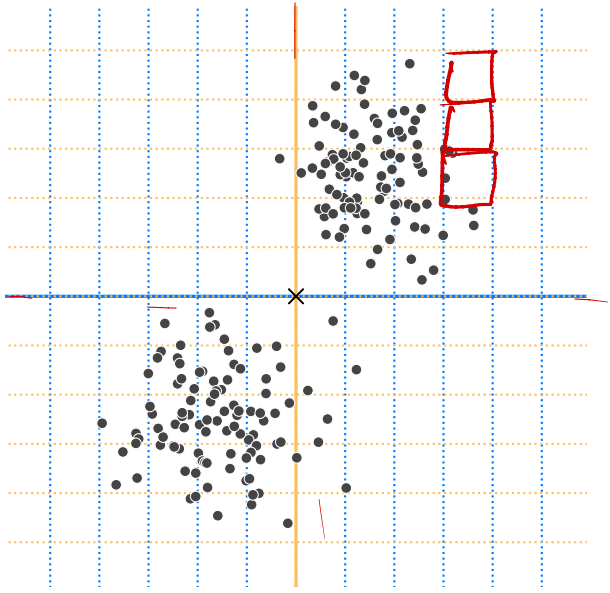
Lecture 5 | Part 8

Implementing a NN Grid

Grids

- Given input point p, want quick way to find nearby points.
- Idea: divide space into cells using grid.
- Find cell containing p, search it.
- How would we implement this?





Grid Cells

- Each point (x, y) given cell id: $(\lfloor x \rfloor, \lfloor y \rfloor)$
  - Example: $(1.2, 6.7)$ given cell id $(1, 6)$.
- Store (x, y) in dictionary with cell id as key.
  - Discretization allows multiple points in same cell.
  - Store collisions in list.
- Generalizes naturally to d-dimensions.

```
class NNGrid:

    def __init__(self, width):
        self.width = width
        self.cells = {}

    def cell_id(self, p):
        p = np.asarray(p)
        cell_id = np.floor(p / self.width).astype(int)
        return tuple(cell_id)

    def insert(self, p):
        """Insert p into the grid."""
        cell_id = self.cell_id(p)
        if cell_id not in self.cells:
            self.cells[cell_id] = []
        self.cells[cell_id].append(p)

    ...
```

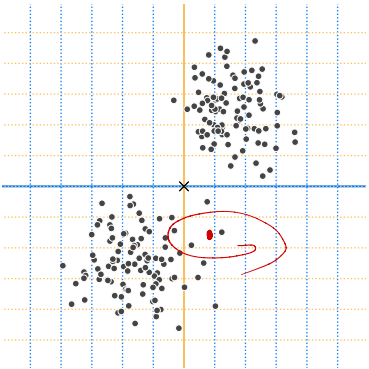
```
...

def points_in_cell(self, p):
    cell_id = self.cell_id(p)
    if cell_id not in self.cells:
        return []
    points_in_cell = self.cells[cell_id]
    # turn into an array
    return np.vstack(points_in_cell)

def query(self, p):
    return brute_force_nn(self.points_in_cell(p), p)
```

Note

- This may **fail** – NN could be in different cell.



Problems

- In d dimensions, cell id has d entries.

$$\text{cell-id}(p) = (\lfloor x_1/w \rfloor, \lfloor x_2/w \rfloor, \ldots, \lfloor x_d/w \rfloor)$$

- All entries must be **exactly** the same for two points to have same cell id.
- This is **very unlikely**. Most cells are empty or contain one point.

High-Dimensional Cuboids

- One “fix”: increase cell width parameter.
- Suppose we want it to be likely that any points within distance r are in same cell.
- Then cell width should be $\approx 2r$.

High-Dimensional Cuboids

- But a d-dimensional cuboid of width $2r$ can contain points at distance $2\sqrt{d}r$ from one another!
- For even modest r, the whole data set is in one cell.

Main Idea

Dividing into a grid of cuboids fails in high dimensions. Either the cells are empty, or contain everything, depending on the width!

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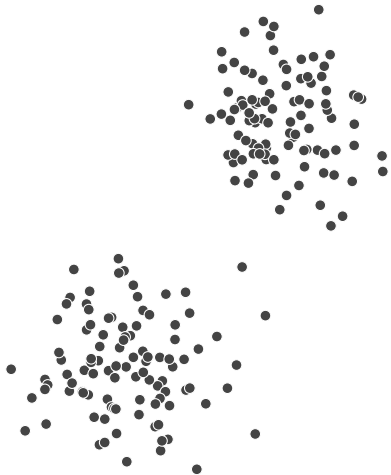
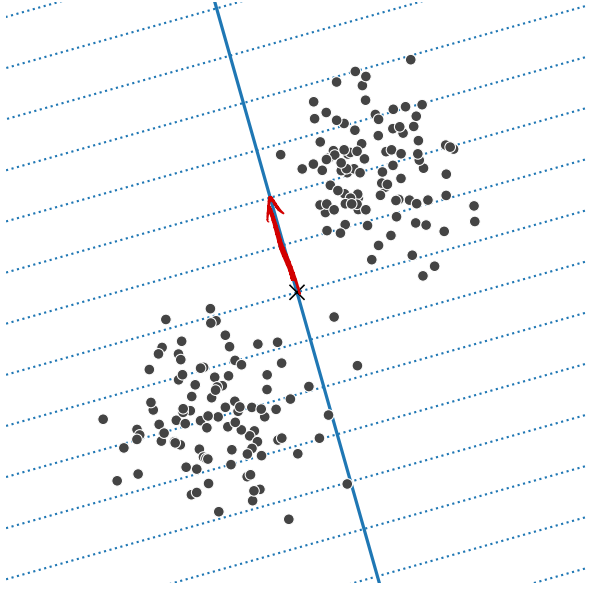
Lecture 5 | Part 9

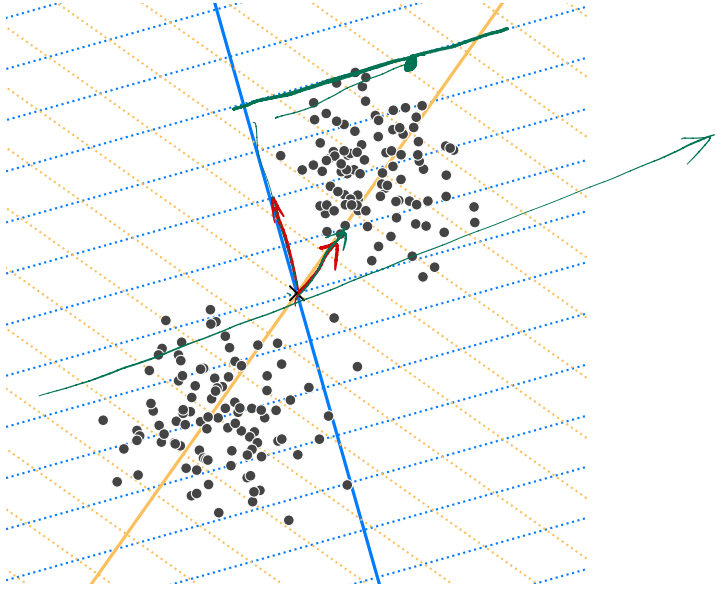
A Randomized “Grid”

A Randomized “Grid”

- Idea: Instead of axis-aligned grid, divide into cells using $k \ll d$ random directions.

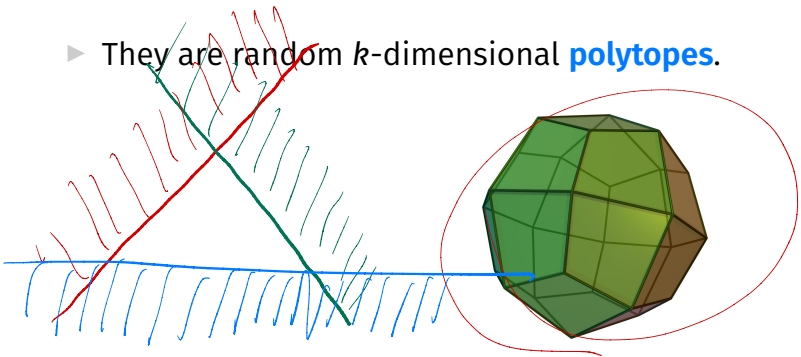






Cell Shape

- Cells are no longer d-dimensional cuboids.
- They are random k-dimensional **polytopes**.

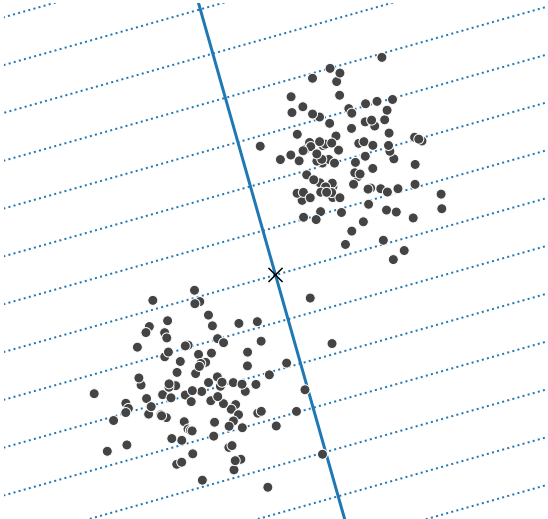


Question

- Why is this better? We'll see in the next sections.

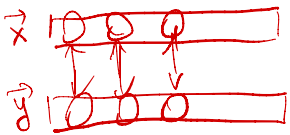
Projection

- How do we determine which cell a point lies in?



Cell IDs

$\vec{x} \cdot \vec{y} = \sum_{i=1}^{d} x_i \cdot y_i$

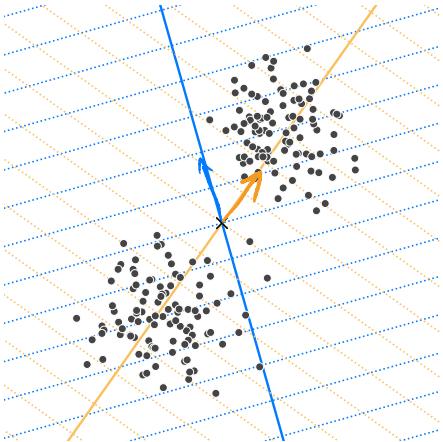


- Pick k random unit vectors, $\vec{u}^{(1)}, \ldots, \vec{u}^{(k)} \in \mathbb{R}^d$.
- Pick a width parameter, w.
- Given any point $\vec{p}$, its cell id is[2]:

$$\text{cell-id}(\vec{p}) = \left(\left\lfloor \frac{\vec{u}^{(1)} \cdot \vec{p}}{w} \right\rfloor, \left\lfloor \frac{\vec{u}^{(2)} \cdot \vec{p}}{w} \right\rfloor, \ldots, \left\lfloor \frac{\vec{u}^{(k)} \cdot \vec{p}}{w} \right\rfloor, \right)$$

[2] use same width and unit vectors for all points

Example



Quick Cell-ID Calculation

- Place $\vec{u}^{(1)}, \ldots, \vec{u}^{(k)}$ into a matrix:

$$U = \begin{pmatrix} \leftarrow & (\vec{u}^{(1)})^T & \rightarrow \\ \leftarrow & (\vec{u}^{(2)})^T & \rightarrow \\ \vdots & \vdots & \vdots \\ \leftarrow & (\vec{u}^{(k)})^T & \rightarrow \end{pmatrix}$$

- Then cell-id($\vec{p}$) = entrywise-floor($U\vec{p}/w$)

Generating Random Unit Vectors

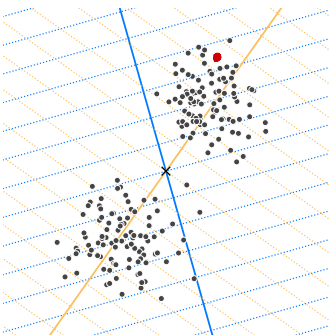
```
def gaussian_projection_matrix(k, d):
    X = np.random.normal(size=(k, d))
    U = X / np.linalg.norm(X, axis=1)[:,None]
    return U
```

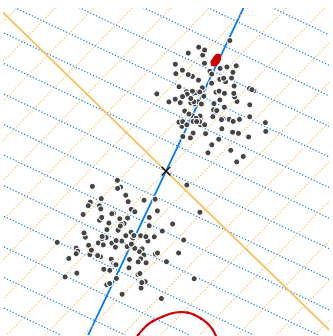
```
class NNProjectionGrid

    def __init__(self, projection_matrix, width):
        self.width = width
        self.projection_matrix = projection_matrix
        self.cells = {}

    def cell_id(self, p):
        projection = self.projection_matrix @ p
        cell_id = np.floor(projection / self.width)
        return tuple(cell_id.astype(int))

    # insert, query, points_in_cell same as for NNGrid
```

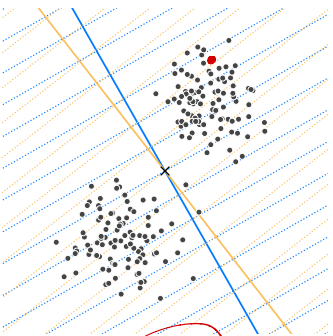
But wait...

- In high dimensions, still **very unlikely** for cell to contain >1 point.
- Idea: **banding**. Try, try again.
- Build multiple `NNProjectionGrids` with different random projections.
- Find `points_in_cell` for each, pool them together.

Multiple Random Projections







Locality Sensitive Hashing

- This idea (multiple random projections) is an example of **Locality Sensitive Hashing** (LSH).
- We'll explore it more in the next section.

```python
class LocalitySensitiveHashing:

    def __init__(self, l, k, d, w):
        self.randomized_grids = []
        for i in range(l):
            U = gaussian_projection_matrix(k, d)
            randomized_grid = NNProjectionGrid(U, w)
            self.randomized_grids.append(randomized_grid)

    def insert(self, p):
        for randomized_grid in self.randomized_grids:
            randomized_grid.insert(p)

    ...
```

```
...

def query_close(self, p):
    nearby = []
    for randomized_grid in self.randomized_grids:
        points_in_cell = randomized_grid.points_in_cell(p)
        nearby.append(points_in_cell)
    return np.vstack(nearby)

def query_nn(self, point):
    results = self.query_close(point)
    pool = np.vstack([r for r in results])
    if len(pool) == 0:
        raise ValueError('No points nearby.')
    return brute_force_nn(pool, point)
```

Parameters

- `l`: number of randomized “grids”
- `k`: number of random directions in each “grid”
- `w`: bin width

Tuning Parameters

- Choose so that `.query_close` returns a small # of points.
- If # is very small (or zero), either:
  - increase w or ℓ
  - decrease k

Note

- This is an approximate NN technique!
- May not find **the** NN.
- May not return **anything**!

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Lecture 5 | Part 10

Theory of Locality Sensitive Hashing

Why does LSH work?

- Two approaches to understanding LSH.
- **1) Hashing view.**
- 2) Dimensionality reduction view.

Standard Hashing

- A **hash function** $h : \mathcal{X} \to \mathbb{Z}$ takes in an object from $\mathcal{X}$ and returns a bucket number.

Standard Hashing

- **Collision**: two different objects have same hash.
- Usually, collisions are **bad**.
- Want similar things to have very different hashes.

Locality Sensitive Hashing

- But in NN search, we want “close” items to be in the **same bucket** (have same hash).
- “Far” items should be in **different buckets** (have different hash).

Locality Sensitive Hashing

- Let r be a distance we consider "close".
- Let cr (with $c > 1$) be a distance we consider "far".
- Suppose H is a **family** of hash functions.

LSH Family

- H is an **LSH family** if when h is randomly drawn from H:

$$\mathbb{P}(h(x) = h(y)) \geq p_1 \quad \text{when } d(x, y) \leq r$$
$$\mathbb{P}(h(x) = h(y)) \leq p_2 \quad \text{when } d(x, y) \geq cr$$

where $p_1 > p_2$.

Main Idea

If x and y are close, the probability that they hash to the **same** bin is not too small. If they are far, the probability is not too large.

Example: Random Projections

- We have seen one LSH family: random projections followed by binning.
- H has infinitely-many hash functions, one for each direction $\vec{u}$:

$$h_{\vec{u}}(\vec{p}) = \left\lfloor \frac{\vec{u} \cdot \vec{p}}{w} \right\rfloor,$$

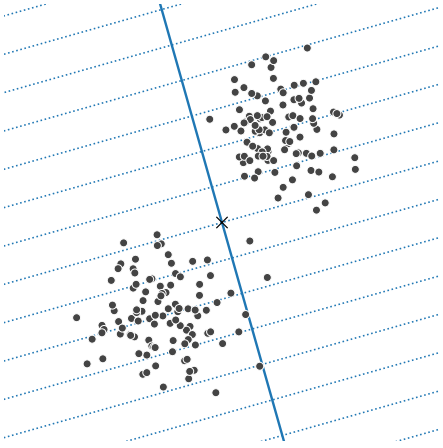
Example: Random Projections

- Suppose a random hash function h is chosen.
- Claim:

$$\mathbb{P}(h(x) = h(y)) \geq \frac{1}{2} \quad \text{when } d(x, y) \leq w/2$$

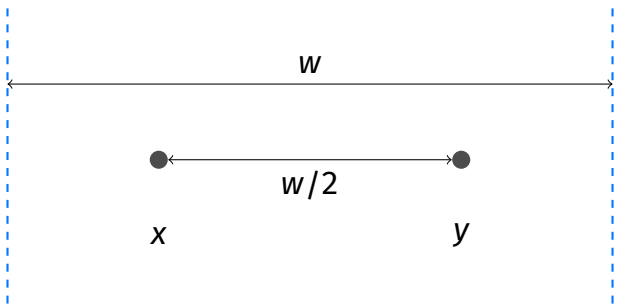
$$\mathbb{P}(h(x) = h(y)) \leq \frac{1}{3} \quad \text{when } d(x, y) \geq 2w$$

Intuition



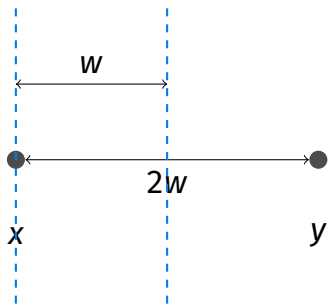
Proof: Close

- In worst case, grid is orthogonal to line between points.



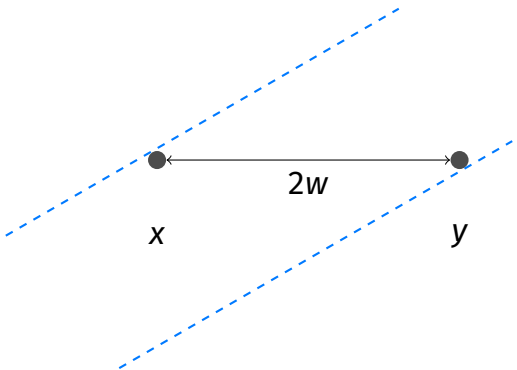
Proof: Far

- Only possible if grid is close to parallel.



Proof: Far

- Angle must be below 30˚.



Amplification

- Lots of points have same hash.
- To be more selective, randomly select k hash functions for cell id.

$$\text{cell-id}(x) = (h_1(x), h_2(x), \ldots, h_k(x))$$

Example: Random Projections

- In case of random projections.

$$\text{cell-id}(\vec{p}) = \left(\underbrace{\left\lfloor \frac{\vec{u}^{(1)} \cdot \vec{p}}{w} \right\rfloor}_{h_1}, \underbrace{\left\lfloor \frac{\vec{u}^{(2)} \cdot \vec{p}}{w} \right\rfloor}_{h_2}, \ldots, \underbrace{\left\lfloor \frac{\vec{u}^{(k)} \cdot \vec{p}}{w} \right\rfloor}_{h_k} \right)$$

Collision Probability

- Remember:

$$P(h(x) = h(y)) \geq p_1 \text{ if close.}$$
$$P(h(x) = h(y)) \leq p_2 \text{ if far.}$$

- Collision occurs if $h_i(x) = h_i(y)\ \forall i \in \{1, \ldots, k\}$.

- Probability of collision...
  - if close: $\geq p_1^k$
  - if far: $\leq p_2^k$

Choosing k

- Want prob. of far points colliding to be small.
- Say, $1/n$.
- Set $p_2^k = 1/n$. Then

$$k = \log_{p_2} \frac{1}{n} = -\frac{\log n}{\log p_2}$$

$k = \Theta(\log n)$

Main Idea

We can use $k = \Theta(\log n)$ hash functions.

Main Idea

When using random projections as hash functions, we can use $k = \Theta(\log n)$ directions. This is usually much less than d.

But wait...

- Probability of close points colliding is p_1^k.
- Let $p_1 = p_2^\rho$. We'll have $\rho < 1$, since $p_2 < p_1$.
- Since $p_2^k = \frac{1}{n}$, we have $p_1^k = \frac{1}{n^\rho}$.
- This is **very small**.

Banding

- Before: one set of k hash functions.
- With **banding**: keep ℓ sets (**bands**) of k hash functions.
- To query NN of p, find points that are in the same cell as p in *any* of the bands.

Banding

- Probability of at least one match:

$$\underbrace{\frac{1}{n^\rho}}_{\text{collision in band 1}} + \underbrace{\frac{1}{n^\rho}}_{\text{collision in band 2}} + \ldots + \underbrace{\frac{1}{n^\rho}}_{\text{collision in band } \ell} = \frac{\ell}{n^\rho}$$

- Want this to be ≈ 1, so:

$$\ell = n^\rho$$

Main Idea

We should set the number of bands to be n^{ρ}. ρ depends on c, and is usually not small. For random projections, $\rho \approx .63$.

Analysis

- How efficient is LSH?
- Worst case, everything hashes to same bin: $O(n)$.
- In practice, much better.
- Requires **a lot** of memory. $\Theta(\ell n)$.

Other Distances

- LSH works for many different similarity measures.
- Random projections are for Euclidean distances.
- But other hashing approaches work for cosine distance, Jaccard distance, etc.

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Lecture 5 | Part 11

The Johnson-Lindenstrauss Lemma

Why does LSH work?

- Two approaches to understanding LSH.
- 1) Hashing view.
- **2) Dimensionality reduction view.**

Main Idea

The **Johnson-Lindenstrauss Lemma** says that, given n points in $\mathbb{R}^d$, you can reduce the dimensionality to $k \approx \log n$ while still preserving relative distances by randomly projecting onto a set of k unit vectors.

Claim

The **Johnson-Lindenstrauss Lemma** (Informal). Let X be a set of n points in $\mathbb{R}^d$. Let U be a matrix whose $k = O(\log(n)/\epsilon^2)$ rows are Gaussian random vectors in $\mathbb{R}^d$. Then for every $\vec{x}, \vec{y} \in X$,

$$\|\vec{x} - \vec{y}\| \leq (1 \pm \epsilon)\|U\vec{x} - U\vec{y}\|$$

LSH and J-L

- In LSH, we use $k = O(\log n)$ hash functions.
- If these hash functions are random projections, the J-L lemma tells that distances are largely preserved.

A Different View of LSH

- Given $p \in \mathbb{R}^d$, randomly project to $\mathbb{R}^k$ with $k \approx \log n$.
- Let new coordinates be $(y_1, y_2, \ldots, y_k)$.
- Use standard grid to assign cell id.

Main Idea

LSH (for Euclidean distances) (without banding) can be viewed as dimensionality reduction by random projections, followed by binning into a standard grid.

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Lecture 5 | Part 12

NN in Practice

In Practice

- LSH is an important idea.
- Good performance in practice.
- But heuristic approaches are often faster.
- `faiss` and `annoy`, among others.

Other Approaches

- Hierarchical k-means.
- Product quantization.
- Navigable small worlds.

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Lecture 5 | Part 13

Disjoint Sets

Disjoint Sets

- Often need to keep a collection of **disjoint sets**.
  - Example: $\{\{4, 6, 2, 0\}, \{1, 3\}, \{5\}\}$
- May need to union disjoint sets.
- May need to check if two items are in same set.

Use Case

- We are given a **stream** of nodes, edges.
- Want to keep track of CCs at every step.
- BFS/DFS take $\Theta(V + E)$ time; efficient to compute CCs once, but then need to recompute.

Use Cases

- Used in Kruskal's algorithm for MST.
- Used in single linkage clustering.
- Used in Tarjan's algorithm to find LCA in a tree.

Disjoint Sets, Abstractly

- A **disjoint sets** ADT represents a collection of disjoint sets.
  - Example: $\{\{4, 6, 2, 0\}, \{1, 3\}, \{5\}\}$
- Supports three operations:
  - `.make_set()`, `.find_set(x)`, `.union(x, y)`
- Sometimes called a **Union-Find** data type.

Assumption

- Elements are consecutive integers.
  - Example: $\{\{4, 6, 2, 0\}, \{1, 3\}, \{5\}\}$
- Not really a limitation.
  - Keep dictionary mapping, e.g., string ids to integers.

.make_set()

- Create a new singleton set.
- Element “id” automatically inferred, returned.

```
>>> ds = DisjointSet()
>>> ds.make_set()
0
>>> ds.make_set()
1
>>> ds.make_set()
2
```

.union(x, y)

- Union sets containing x and y.
- Updates data structure in-place.

```
>>> ds = DisjointSet()
>>> ds.make_set()
0
>>> ds.make_set()
1
>>> ds.make_set()
2
>>> ds.union(0, 2)
```

.find_set(x)

- Find **representative** of set containing x.
- Representative is arbitrary, but same for all items in same set.
- Used to test if two nodes in same set.
- Guaranteed to not change unless a `union` is performed.

```
>>> # ds is {{0}, {1}, {2}}
>>> ds.union(0, 2)
>>> ds.find_set(0)
0
>>> ds.find_set(2)
0
>>> ds.union(0, 1)
>>> ds.find_set(0)
1
>>> ds.find_set(1)
1
>>> ds.find_set(2)
1
```

Today's Lecture

- How do we implement a disjoint set?
- We'll introduce the **disjoint set forest** data structure.
- Talk about two heuristics that make it very efficient.

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Lecture 5 | Part 14

Disjoint Set Forests

Implementing Disjoint Sets

- First idea: a `list` of `set`s.

```
[{2, 4, 3}, {1, 5}, {0}]
```

- **Problem**: unioning two `set`s takes time linear in size of smaller.

Looking Ahead

- We'll design data structure so that all operations, including `union`, take (practically) $\Theta(1)$ time.

The Idea

- Represent collection as a forest of trees, called a **Disjoint Set Forest**.
- Example: $\{\{2, 4, 3, 6\}, \{1, 5\}, \{0\}\}$
- Not unique!

Tree Structure

- Each node has reference to **parent**.
- Not a binary tree!

Representing Forests

- We have several choices:
- 1) Each node is own **object** with `parent` attribute.
- 2) Keep a **list** containing parent of each element.

Approach #1

```
class DSFNode:

    def __init__(self, parent=None):
        self.parent = parent
```

- `make_set` becomes `DSFNode()`
- `find_set` and `union` are functions, not methods.
- They accept `DSFNode` objects.

Assumption

- Now, assume that the elements of the disjoint set are **consecutive integers** $0, 1, \ldots, n - 1$.
- We can store the tree in a single list of size n.
- `arr[i]` is parent of element i:

```
[2, 0, None, 2, 4]
```

Approach #2

```
class DisjointSetForest:

    def __init__(self):
        # self._parent[i] is
        # parent of element i
        self._parent = []

    def make_set(self):
        ...

    def find_set(self, x):
        ...

    def union(self, x, y):
        ...
```

Implementation Notes

- We'll use the second approach.
- We can use second representation because elements are consecutive integers.
- For cache locality, use `numpy` array, not `list`.

.make_set

```
def make_set(self):
    # infer new element's "id"
    x = len(self._parent)
    self._parent.append(None)
    return x
```

```
>>> dsf = DisjointSetForest()
>>> dsf.make_set()
0
>>> dsf.make_set()
1
>>> dsf.make_set()
2
>>> dsf._parent
[None, None, None]
```

`.find_set(x)`

- Idea: use the “root” as the representative.

Exercise

Implement `.find_set(x)` recursively.

.find_set

```
def find_set(self, x):
    if self._parent[x] is None:
        return x
    else:
        return self.find_set(self._parent[x])
```

`.union(x, y)`

- Idea: make one root the parent of the other.

.union(x, y)

```
def union(self, x, y):
    x_rep = self.find_set(x)
    y_rep = self.find_set(y)
    if x_rep != y_rep:
        self._parent[y_rep] = x_rep
```

```
>>> # dsf is {{0}, {1}, {2}}
>>> dsf._parent
[None, None, None]
>>> dsf.union(0, 1)
>>> dsf._parent
[None, 0, None]
>>> dsf.union(1, 2)
>>> dsf._parent
[None, 0, 0]
```

Analysis

- `.make_set`: $\Theta(1)$ time[3]
- `.union`: depends on `.find_set`
- `.find_set`: $O(h)$, where h is height of tree

[3] Amortized, since we're using a dynamic array. But truly $\Theta(1)$ with an over-allocated static array or in the object representation.

Tree Height

- Trees can be very deep, with $h = O(n)$.
  - `.find_set` and `.union` can take $\Theta(n)$ time!

- Example:

```
# dsf is {{0}, {1}, {2}, {3}, {4}}
>>> dsf.union(1, 0)
>>> dsf.union(2, 1)
>>> dsf.union(3, 2)
>>> dsf.union(4, 3)
```

Tree Height

- But trees can also be shallow, with $h = O(1)$.

- Example:

```
# dsf is {{0}, {1}, {2}, {3}, {4}}
>>> dsf.union(0, 1)
>>> dsf.union(1, 2)
>>> dsf.union(2, 3)
>>> dsf.union(3, 4)
```

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Design and Analysis of Algorithms I

Lecture 5 | Part 15

Path Compression and Union-by-Rank

The Bad News

- We saw that the tree can become very deep.
- In worst case, `.find_set` and thus `.union` take $\Theta(n)$ time.

Heuristics

- Now: two heuristics helping trees stay shallow.
- **Union-by-Rank** and **Path Compression**
- Together, these result in a **massive** speed up.

Path Compression

- Idea: if we find a long path during `.find_set`, “compress” it to (possibly) reduce height.

.find_set

```
def find_set(self, x):
    if self._parent[x] is None:
        return x
    else:
        root = self.find_set(self._parent[x])
        self._parent[x] = root
        return root
```

Union-by-Rank

- Should we `.union(x, y)` or `.union(y, x)`?

Union-by-Rank

- Placing deeper tree under shallower tree increases height by one.
- But placing shallower tree under deeper tree doesn't increase height.
- **Idea**: always place shallower tree under deeper.

Rank

- We need to keep track of height (**rank**) of each tree.
- Store `rank` attribute.
- `rank[i]` is height[4] of tree rooted at node i.

[4] Exactly the height if path compression isn't used, but upper bound if it is.

Rank

```
class DisjointSetForest:

    def __init__(self):
        self._parent = []
        self._rank = []

    def make_set(self):
        # infer new element's ”id”
        x = len(self._parent)
        self._parent.append(None)
        self._rank.append(0)
        return x
```

.union

```
def union(self, x, y):
    x_rep = self.find_set(x)
    y_rep = self.find_set(y)

    if x_rep == y_rep:
        return

    if self._rank[x_rep] > self._rank[y_rep]:
        self._parent[y_rep] = x_rep
    else:
        self._parent[x_rep] = y_rep
        if self._rank[x_rep] == self._rank[y_rep]:
            self._rank[y_rep] += 1
```

Note

- With path compression, rank is no longer *exactly* the height – it is an upper bound.
- But this is good enough.

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Lecture 5 | Part 16

Analysis

Analysis of DSF

- A DSF with path compression and union-by-rank ensures trees are shallow.
- How does this affect runtime?

Answer

- Assuming union-by-rank and path compression...
- In a sequence of m operations, n of which are `.make_sets`...
- Amortized cost of a single operation is $O(\alpha(n))$.
- α is the **inverse Ackermann function**, and it is essentially constant.

Inverse Ackermann

$\alpha(n)$	n
0	$n \in [0, 1, 2]$
1	$n = 3$
2	$n \in [4, \ldots, 7]$
3	$n \in [8, \ldots, 2047]$
4	$n \in [2048, \ldots, 2^{2048}]$ and beyond

Proof

- The formal analysis is quite involved.
- But we'll provide some intuition.

Union-by-rank Alone

- Union-by-rank alone ensures height is $O(\log n)$.

```
# dsf is {{0}, {1}, {2}, {3}}
>>> dsf.union(0, 1)
>>> dsf.union(2, 3)
>>> dsf.union(0, 2)
```

Union-by-rank Alone

- Union-by-rank alone ensures `.find_set` is $O(\log n)$.

Path Compression + U-by-R

- With path compression, individual `.find_set` calls can take $O(\log n)$.
- But they massively improve subsequent calls.
  - For other nodes, too!